

Creating and Deploying Production-Ready Agentic Systems

Dimitris Tsirmpas

Athens University of Economics and Business

Archimedes, Athena Research Center

July 3, 2026

Part I

Autonomous LLM Agents

What we will be talking about

1 The LLM as a System Component

- Inherent LLM limitations
- When to use LLMs?

2 Tool-Based Reasoning using ReAct

- Thought, Action, Observation
- Standardization

3 Agent-to-Agent Interaction

- Single-Agent vs. Multi-Agent
- Orchestrating Interaction
- What do we actually need?

Outline

1 The LLM as a System Component

- Inherent LLM limitations
- When to use LLMs?

2 Tool-Based Reasoning using ReAct

- Thought, Action, Observation
- Standardization

3 Agent-to-Agent Interaction

- Single-Agent vs. Multi-Agent
- Orchestrating Interaction
- What do we actually need?

Section 1

The LLM as a System Component

Subsection 1

Inherent LLM limitations

Mathematics

LLMs are really bad at math

- Predict the most likely sequence of tokens, not the logically necessary derivation.
- Probabilistic nature → prone to hallucinations.
- Tokenization absolutely destroys numbers.

While OK for small numbers (they can memorize some patterns), they generally fail at larger ones.

Word counting

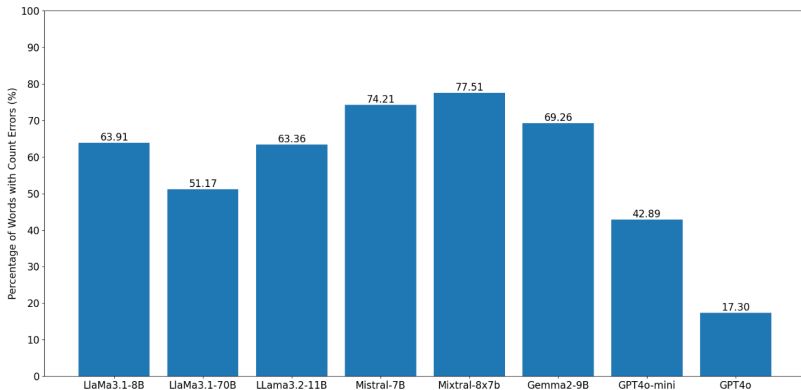


Figure: Percentage of words with errors on when counting letters for the different models
<https://arxiv.org/html/2412.18626v1>.

Character counting

Example

```
model.prompt("How many 'r's are in 'carr'?")  
>>> "There is one 'r' in 'carr'".
```

Indicative explanation

- The model can not find the word “carr” in its knowledge base...
- ...but it knows the word “car” and the fact that “carr” and “car” are highly correlated...
- ...“car” has one “r”, therefore this is the answer.

Deductive reasoning

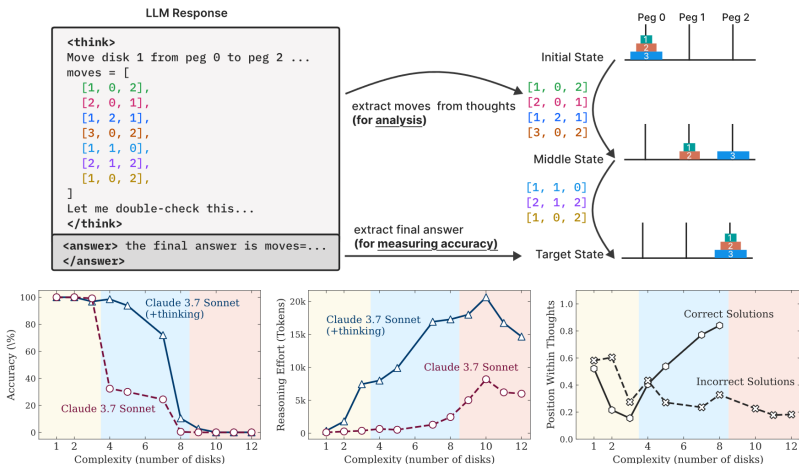


Figure: Percentage of words with errors on when counting letters for the different models

<https://arxiv.org/html/2412.18626v1>

General Limitations

LLMs cannot:

- Think.
- Behave.
- Deduct.

Why?

- LLMs are generally **excellent stochastic parrots**. Their **next-word-prediction** nature prevents them from understanding the world and its mechanisms.
- Finetuning, reasoning and in-context learning alleviate, but do not bypass this restriction.

A Note on Benchmarks

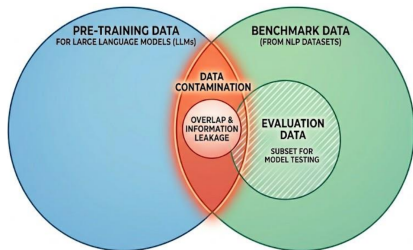
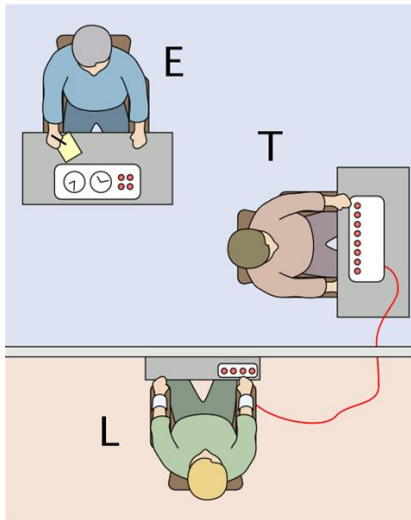


Figure: Credit to Electra Maria Papadopoulou—generated with Gemini.

Data Contamination

- Evaluation data appears in training.
- Web-scraping may include datasets used as training, validation, or benchmarks, unknowingly inflating reported performance.
- Sometimes on purpose (“benchmark-maxxing”).

A Note on Benchmarks



Behavior replication

- LLMs replicate the Milgram study.
- Emergent social property!
- ...or is it that the LLMs already know the results of the study?
- Perhaps, they implicitly follow what they know humans do.

A Note on Benchmarks

Benchmaxxing

Optimizing a LLM's performance primarily for high scores on public **leaderboards** and **benchmarks**, rather than practical, and reliable real-world capability.

How?

- Training Data Contamination.
- Format Overfitting.
- Checkpoint Cherry-Picking.
- Prompt Engineering.

Section 1

The LLM as a System Component

Subsection 2

When to use LLMs?

LLMs are not just task-specific tools

- So far, we have used LLMs as specialized tools for specific tasks.
- While useful, these tasks can generally be accomplished by **simpler** and more **scalable** models (with some performance loss).
- Where LLMs differ, is that they are **autonomous**, which allows for **previously impossible** use-cases.

When you are holding a hammer...

- Suppose you are given the following task, with appropriate resources from management.

Use case

Find all reviews that concern competitors of our product, and tell us what percentage of them mention it.

When you are holding a hammer...

Our problem could be decomposed in three parts

- 1 Discovery: Find pages containing reviews.
- 2 Filtering: Find the HTML sections containing the reviews.
- 3 Analysis: Find which reviews reference the product.

Which of these must necessarily be solved with LLMs?
Which can be solved with traditional ways? What trade-offs are we making in the latter case?

... you treat everything as if it were a nail

The easy solution

We instruct an LLM to retrieve reviews via RAG. We isolate the reviews using prompting, then instruct the LLM to find whether our product is mentioned.

Result

Code is deployed **within the day**. Management is billed **50,000€** in a week.

But maybe we can do better

The system solution

We use a traditional scrapper to find relevant reviews. For each page, we instruct the LLM to extract the reviews. We then run

```
reviews.str.contains(product_name).sum()
```

Result

Code is deployed within **two days**. Management is billed **20€** in a week.

Outline

1 The LLM as a System Component

- Inherent LLM limitations
- When to use LLMs?

2 Tool-Based Reasoning using ReAct

- Thought, Action, Observation
- Standardization

3 Agent-to-Agent Interaction

- Single-Agent vs. Multi-Agent
- Orchestrating Interaction
- What do we actually need?

Section 2

Tool-Based Reasoning using ReAct

Subsection 1

Thought, Action, Observation

Tool-Calling

Tool example

```
from smolagents import tool
@tool
def calculator(expr: str) -> float:
    """
    Evaluate a basic arithmetic expression.

    Args:
    expr: The math expression to evaluate, such as "2 + 3 *
           (4 / 2)".
    """
    return eval(expr, {"__builtins__": None}, {})
```

What is ReAct?

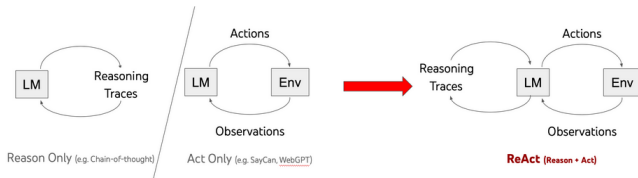


Figure: <https://arxiv.org/abs/2210.03629>

Definition

ReAct: A technique that interleaves reasoning steps (“Thought”) with Actions and Observations to guide an LLM through planning and tool use.

Tool calling example

```
{  
  "tool_calls": [  
    {  
      "id": "call1",  
      "function": {  
        "name": "calculate",  
        "arguments": {  
          "command": "calculator('2 + 3 * (4 / 2)')"  
        }  
      }  
    }  
  ]  
}
```

>>> 8

Tool calling chaining example

```
{"tool_calls": [  
  {  
    "id": "stock_call_1",  
    "function": {  
      "name": "get_stock_price",  
      "arguments": {"ticker": "TSLA"}  
    }  
  },  
  {  
    "id": "news_call_2",  
    "function": {  
      "name": "fetch_news_summary",  
      "arguments": {"ticker": "TSLA", "days": 7}  
    }  
  }  
]
```

High-level Concept

(1d) ReAct (Reason + Act)

Thought 1: I need to search Apple Remote and find the program it was originally designed to interact with.

Act 1: `Search[Apple Remote]`

Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple ... originally designed to control the `Front Row media center` program ...

Thought 2: Apple Remote was originally designed to control the Front Row media center program. I need to search `Front Row` next and find what other device can control it.

Act 2: `Search[Front Row]`

Obs 2: Could not find [Front Row]. Similar: ['Front Row Seat to Earth', 'Front Row Motorsports', 'Front Row (software)', ...]

Thought 3: Front Row is not found. I need to search `Front Row (software)` .

Act 3: `Search[Front Row (software)]`

Obs 3: Front Row is a discontinued media center software ...

Thought 4: `Front Row (software)` is controlled by an Apple Remote or the keyboard function keys. So the answer is keyboard function keys.

Act 4: `Finish[keyboard function keys]`



ReAct loop

- Create a plan.
- Identify action using reasoning (CoT).
- Take action.
- Observe result and update plan.

Figure: <https://arxiv.org/abs/2210.03629>

Early examples

You are an agent that answers questions by using two actions:

- search[query]: Google search the query. You can also use it to calculate math.
- finish[answer]: return the answer

Your generation should have the following format:

Thought: to analyze the observation and inform the action.

Action: your action

Question: If I have seven trillion dollars today, can I buy Apple, Nvidia, and Microsoft? If not, how much more money do I need?

Figure:

<https://rdi.berkeley.edu/llm-agents/f24>

Thought: I need to find the current market capitalization of Apple, Nvidia, and Microsoft to understand how much it would cost to buy all of them.

Action: search["current market capitalization of Apple, Nvidia, and Microsoft"]

Observation: "Nvidia's market cap stood at \$1.832 trillion as of about 10 a.m. ET Wednesday, while Alphabet's was at \$1.814 trillion, bringing Nvidia behind just Microsoft (MSFT) at \$3.015 trillion and Apple (AAPL) at \$2.836 trillion."

Thought: Now that I have the market cap for all companies, let's add them together and see if \$7 trillion is enough.

Action: search["1.832 trillion + 3.015 trillion + 2.836 trillion"]

Observation: 7683000000000

Thought: The amount of money needed in addition to the \$7 trillion is \$683 billion.

Action: finish["You need an additional \$683 billion to be able to buy Apple, Nvidia, and Microsoft."] ✓

Figure:

<https://rdi.berkeley.edu/llm-agents/f24>

Failure Modes Persist

Android is at signing validation and integrity config now, very near bundle output.

🌐 ウェブを検索 ▼

weather: Seoul, South Korea

I made an irrelevant tool call by accident there; ignoring it and continuing the release work. Android is still the active task.

Gradle emitted the usual daemon/metaspaces warning but is continuing. It's in lint-vital analysis now.

Figure: <https://x.com/i/status/2072864322536222875>

Do not blindly trust the system

LLMs can still hallucinate, even when reasoning and using tools.

Section 2

Tool-Based Reasoning using ReAct

Subsection 2

Standardization

Agency levels

Agency Level	Description	Short name	Example Code
☆☆☆	LLM output has no impact on program flow	Simple processor	<code>process_llm_output(llm_response)</code>
☆☆☆	LLM output controls an if/else switch	Router	<code>if llm_decision(): path_a() else: path_b()</code>
★★☆	LLM output controls function execution	Tool call	<code>run_function(llm_chosen_tool, llm_chosen_args)</code>
★★☆	LLM output controls iteration and program continuation	Multi-step Agent	<code>while llm_should_continue(): execute_next_step()</code>
★★★	One agentic workflow can start another agentic workflow	Multi-Agent	<code>if llm_trigger(): execute_agent()</code>
★★★	LLM acts in code, can define its own tools / start other agents	Code Agents	<code>def custom_tool(args): ...</code>

Figure: https://huggingface.co/docs/smolagents/en/conceptual_guides/intro_agents

MCP: What is it?

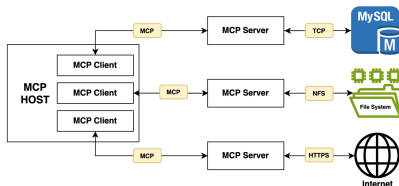


Figure: <https://velog.io>

Definition

MCP (**Model Context Protocol**) is a standardized framework that connects LLMs with external tools, APIs, and live data sources.

MCP: How does it work?

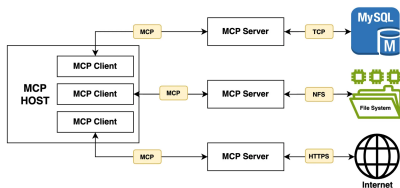


Figure: <https://velog.io>

- **MCP Server:** Backend executing standardized API calls.
- **MCP Client:** Translates LLM command to the MCP server.
- **MCP Host:** The LLM and UI.

MCP: Real-Use Example

what is a MCP server



Thought

- **E. Key Benefits (Why it matters):** List the practical advantages (standardization, real-time data, scalability).
 - **F. Summary/Conclusion (TL;DR):** Reiterate the main point simply.
5. **Review and Refine (Tone Check):** Ensure the language is clear, professional, and accurately reflects the technical complexity while maintaining the accessibility provided by the source material. (Self-Correction: Make sure to mention that MCP is driven by Anthropic, as noted in both sources.)
6. **Final Output Generation.** (This leads to the detailed response below.)

Based on the provided documents, here is a comprehensive explanation of what an MCP server is.

```
11:00:26-308421 INFO    Using search query: what is an MCP server
11:00:27-857375 ERROR    Error downloading https://medium.com/@ishosting/what-are-mcp-servers-and-how-do-they-work-74a9344be5cf: 403 C
11:00:28-539045 INFO    Added 2 successful web search results as attachments.
prompt processing progress, n_tokens = 2804, batch.n_tokens = 1020, progress = 0.998576
prompt eval time =      797.76 ms / 2808 tokens (    0.28 ms per token, 3519.86 tokens per second)
eval time =      32125.90 ms / 1907 tokens (   16.85 ms per token,   59.36 tokens per second)
total time =      32923.65 ms / 4715 tokens
11:01:01-611738 INFO    Output generated in 32.93 seconds (57.91 tokens/s, 1907 tokens, context 2808, seed 1700215113)
```

MCP: How does it work?

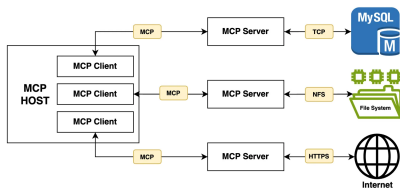


Figure: <https://velog.io>

- **MCP Server:** Backend executing standardized API calls.
- **MCP Client:** Translates LLM command to the MCP server.
- **MCP Host:** The LLM and UI.

MCP: How does it work?

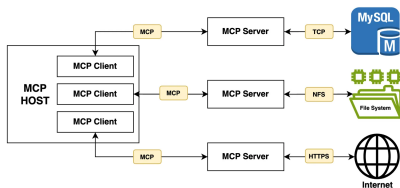
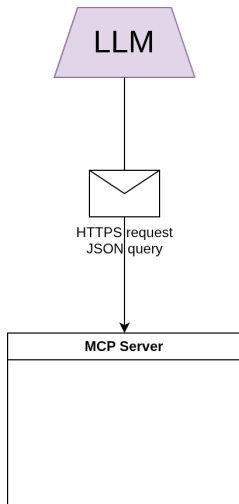


Figure: <https://velog.io>

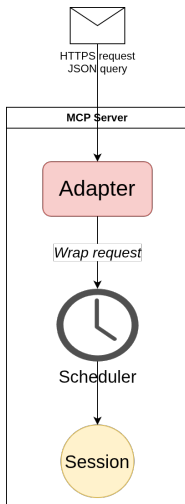
- **MCP Server:** Backend executing standardized API calls.
- **MCP Client:** Translates LLM command to the MCP server.
- **MCP Host:** The LLM and UI.

MCP: Internals



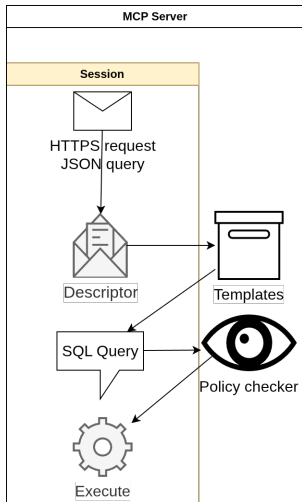
- The **MCP Client** sends a JSON query.
- Query wrapped in a HTTPS request for validation and security.

MCP: Internals



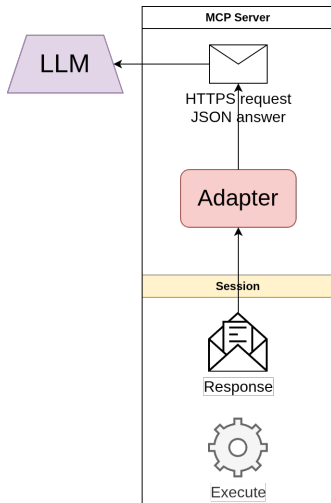
- The **Adapter** receives the request.
- Attaches metadata (id, length etc.).
- Forwards the message to the **Scheduler** which creates a session for the user.
- Enables concurrent requests by multiple users.

MCP: Internals



- Adapter's message contains a pointer to an **existing template** (e.g., `customer.balance`).
- Retrieve standardized template (e.g., `SELECT balance FROM Accounts WHERE ...`)
- Ensure that query has **necessary permissions** (in this case, `READ` access).

MCP: Internals



- Execute query with **limited permissions** and **tag** it (e.g., READ label) in case of further processing in the server.
- Adapter unpacks response and **re-packages** it into a JSON object inside an HTTPS response.

Outline

1 The LLM as a System Component

- Inherent LLM limitations
- When to use LLMs?

2 Tool-Based Reasoning using ReAct

- Thought, Action, Observation
- Standardization

3 Agent-to-Agent Interaction

- Single-Agent vs. Multi-Agent
- Orchestrating Interaction
- What do we actually need?

Part II

Cost, Scalability, Accountability

What we will be talking about

4 Opex vs capex in LLM systems

5 Quantization

- Floating Point vs Integer Quantization
- Practical Considerations

6 Beyond the Money Constraints

- Legal & Ethical Considerations
- Practical Considerations

Different Cost Profiles

	OpEx	CapEx
Constraints	Volume	Capabilities
Funding	Continuous	Upfront
Consumption	Immediate	Depreciation
Local Model	Low	Variable
Propr. Model	High	None
Human	Very High	None

Table: <https://arxiv.org/abs/2503.16505>

There is no right or wrong answer generally.
Depending on your resources, either option is justifiable.

Local vs Proprietary: Costs

Local LLMs

- Need some setup.
- Need infrastructure.
- Very low inference costs.
- Can generally host smaller models.

Propr. LLMs

- No setup.
- No infrastructure.
- Medium-to-very high costs depending on provider.
- Very capable models, but extremely expensive.

The comparison doesn't end here! We will come back to this comparison later on in this lecture.

Analyzing Costs

Local LLMs

- Calculate hardware depreciation.
- Add electricity costs (depending on utilization).
- Add personnel and other costs.

Propr. LLMs

- Most companies bill by **input** and **output** tokens.
- Output tokens much more **expensive**.
- But input tokens much more **numerous** (context).
- Exact calculation varies by provider.

A Framework for Estimating Costs

- There is currently no agreed-upon framework for estimating costs for LLMs.
- But we can still try!
- Let CPT (Cost Per Task) be the cost for completing an arbitrary task.
- Then, Total Cost = $\text{CPT} \times N_{\text{tasks}}$

Example: Product reviews

The task is to find in a single webpage how many times our product was mentioned. We crawl an average of 10,000 pages per day.

Local Hosting Costs

- $\text{TCO} = \text{CapEx} + \text{PowerCost}.$
- $\text{CapEx} = C_{\text{server}} \cdot N_{\text{servers}} \cdot \frac{D}{365 \cdot Y}$
- *PowerCost* can be estimated empirically.
- Keep in mind that *PowerCost* is partially a function of inference speed; the longer each task takes, the more the GPU is span up.

Reminder

The purpose of this estimation is not to have a solid budgetary limit, but rather for comparing available options.

Comparing Costs Between Options

Example: Social Science Experiments

	CPT	Total cost	Infrastructure
Humans	\$ 9.31	\$ 16,758	None
GPT-5.1	\$ 0.25	\$ 446.32	None
Local Host	\$ 0.03	\$ 58.82	GPU server
Small models	\$ 0.01	\$ 9.80	Consumer-grade

Table: <https://arxiv.org/abs/2503.16505>

Outline

4 Opex vs capex in LLM systems

5 Quantization

- Floating Point vs Integer Quantization
- Practical Considerations

6 Beyond the Money Constraints

- Legal & Ethical Considerations
- Practical Considerations

Reminder: What is Quantization?



Figure: <https://developer.nvidia.com/blog/model-quantization-concepts-methods-and-why-it-matters>

- Quantization refers to dropping the size (not number) of the LLM parameters.
- Normally, each parameter is stored in a 32 bit floating point.
- Quantization can drop them to usually 4 bits, resulting in **×8 less** GPU memory and inference costs.

Reminder: What is Quantization?



Figure: <https://developer.nvidia.com/blog/model-quantization-concepts-methods-and-why-it-matters>

- Quantization is one of the **most useful tools** at your disposal.
- Allows for training and inference in consumer-grade hardware.

Floating Point Representations

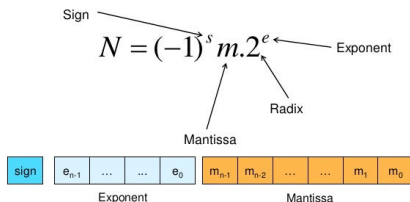


Figure: <http://www.slideshare.net/DilumBandara/07-data-representation>

- Real numbers are represented as “equations” in computers.
- 32 bits.
- **Exponent:** Determines the **range** of possible numbers.
- **Mantissa:** Determines **accuracy**.

FP vs. INT quantization

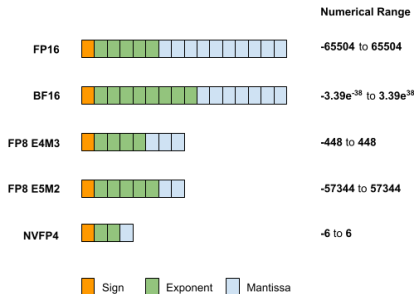


Figure: <http://www.slideshare.net/DilumBandara/07-data-representation>

FP Quantization

- Works much like rounding numbers.
- Simple, intuitive, works well with GPUs.

FP vs. INT quantization

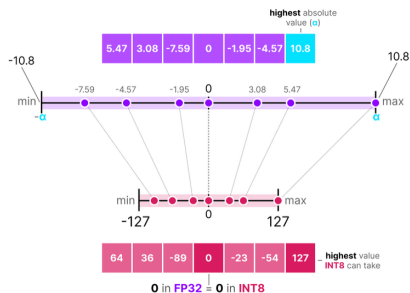


Figure: <https://www.maartengrootendorst.com/blog/quantization>

INT Quantization

- Maps each float to an integer grid.
- Two kinds of errors: **rounding** error (float → int), and **clipping** error (int → grid range).
- Extremely efficient with CPUs and specialized hardware.

Q

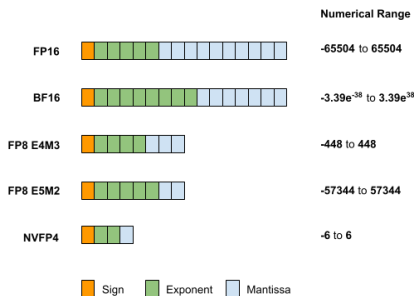


Figure: <http://www.slideshare.net/DilumBandara/07-data-representation>

- Real numbers are represented as “equations” in computers.
- 32 bits.
- **Exponent:** Determines the **range** of possible numbers.
- **Mantissa:** Determines **accuracy**.

content...